Open TechEd: Revival of Legacy Teaching Instruments for Modern Education

Bringing >€200,000 of Laboratory Instruments Back to Life

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Your lab owns the hardware. The company owns the software. And when the software stops, the whole instrument stops.

When a manufacturer declares an instrument "legacy," it stops updating the software that controls it. The hardware still works perfectly, but it can no longer run on modern computers. Result: working instruments get thrown away, not because they broke, but because the software that ran them was abandoned.

The scale of the problem, in numbers: ten Thermo Fisher NanoDrop 1000 spectrophotometers bought in 2004 cost about €200,000. The last software update came in 2011; from 2012 (Windows 8) the device could no longer communicate with modern systems. Eight years after purchase, a €200,000 investment was unusable, purely because of software.

What we did

We picked one such "retired" NanoDrop and revived it. By analysing the data it sends over USB, we reverse-engineered how it works and wrote our own open-source Python software to control it. We then built it into PyLabRobot, an open platform for laboratory automation.

The revived instrument now runs on Windows, macOS and Linux, and its measurements were verified: the light spectrum matches the manufacturer's reference, and it correctly detected a real substance (paracetamol) in water. All code is free and open.

What it means for WUR

Cost savings. Instead of replacing working hardware, we extend its life at almost zero cost. The revived instrument needs no license, no subscription, no vendor key. A free and open piece of software replaces the outdated package that turned a >€200,000 asset worthless.

Openness. Full ownership of both instrument and data. The code is open-source and shared, anyone at WUR (or beyond) can use, inspect and improve it. No hidden costs, no vendor lock-in.

Sustainability. We keep fully functional hardware in use and out of the waste stream, directly supporting WUR's sustainability goals by reducing laboratory e-waste.

Digital sovereignty. Our instrument no longer depends on a vendor that can stop support, lock the device remotely, or hold our data in the cloud. It works on any modern machine, on our own terms.

The opportunity

This pilot proves the approach works on a real instrument. Scaling it to the many other "legacy" instruments across WUR turns old, unused assets back into usable ones, saving money, cutting e-waste, and giving the university full control of its own equipment.

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